

Bibin Thomas

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Professional Summary

Embedded Systems Engineer with an **MS in Computer Science, Be in Electronics and communication engineering** and a strong background in hardware-software integration, robotics, and microcontroller development. Adept at bridging the gap between low-level firmware and high-level applications, utilizing core proficiencies in **Embedded C, C++, and Python** alongside hands-on experience with **STM32, ESP32, and ARM Cortex M4** architectures. Proven track record of optimizing motion control algorithms, implementing robust communication protocols (**UART, SPI, I2C**), and developing end-to-end robotics projects that enhance system efficiency and mobility. Currently expanding expertise in advanced **embedded architectures, driver development, and RTOS**, bringing a multidisciplinary mindset to high-impact engineering teams

Certifications

Embedded systm software engineering | Entri Elevate (present)

- Trained in **embedded c programing, firmware development and RTOS** integration
- Develop drivers and optimize firmware across industry lead microcontrollers
- Design and debug real-time. low power and AI- intergrated embedded systems

Full-Stack Web Development | Unified Mentor (Nov 2025 – Feb 2026)

- Engineered scalable web architectures, translating design concepts into functional **HTML/CSS/JavaScript** components.
- Developed specialized admin and user modules with role-based access control and integrated **Chart.js** for interactive analytics dashboards
- **Expense tracker** - https://github.com/Bbn047/Expense_Tracker
- **Gym management system** - https://github.com/Bbn047/GYM_Management_System
- **Electric vehicle recharge** - <https://github.com/Bbn047/Electric-Vehicle-Recharge-Bunk>

Python Programming | Tech U Tech Pvt Ltd (2021)

- Python Full-Stack training program focusing on modern web development frameworks, database management, and server-side logic.

Training on Piccolo Microcontroller | MC Technologies (2020)

- Gained hands-on experience programming and configuring the **TI Piccolo microcontroller** for real-time control applications and hardware interfacing.

Projects

Audio guidance system for the visually impaired

- Computer Vision & AI: Implemented real-time object detection using **Python, YOLOv5, and OpenCV**, achieving 95% accuracy in recognizing obstacles and environmental hazards.
- Interface Development: Built a web-based companion dashboard using **HTML, CSS, and JavaScript** to visualize live data and system diagnostics from the wearable unit.
- Accessibility Features: Integrated a **Text-to-Speech API** to provide real-time audio guidance and used Arduino for hardware-to-web data synchronization.

Evacuation Vehicle for Military Applications

- Designed and developed a wireless-controlled autonomous evacuation vehicle for military operations, reducing response time in war zones by 40%.
- Developed a robotic arm for assisting injured individuals, and clearing obstacles.
- Implemented a targeting system with 90% accuracy to locate injured personnel within a radius of 50 meters and detect enemy threats.
- Utilized **Arduino Nano** and **NRF24L01** modules, ensuring real-time wireless communication.
- Programmed functions using **C** and **Python**, optimizing system efficiency and reducing processing time by 30%.

Hand writing Robot

- Designed & developed a robot capable of replicating human handwriting with $\pm 0.5\text{mm}$ accuracy.
- Implemented PID control algorithms in C/C++ for precise motor movement.
- Utilized Arduino Nano, CNC Shield V4, and A4988 stepper motor drivers to achieve step resolution of 1/16 microsteps.
- Integrated two NEMA 17 stepper motors (1.8° step angle, 12V, 1.5A) for X-Y axis movement and a servo motor (SG90, 4.8V-6V, 2.5kg.cm torque) for pen control.
- System calibrated for multiple font sizes (5mm to 30mm height) and different writing styles with 95% accuracy in character reproduction.
- Successfully tested with over 50 handwritten samples, demonstrating consistent performance across various surfaces.

Skills

- **programming Languages:** Python, Embedded C, C++ , HTML, Css
- **Microcontrollers:** STM32, ESP32, ARM Cortex M4, TI piccolo microcontroller
- **Protocols:** UART, SPI, I2C
- **Tools:** Git/GitHub, VS Code, Pycharm, Figma, AutoCAD, Blender, Dev C++, Code Composer Studio
- **Web Integration:** JavaScript, React, Firebase

Work Experience

Junior Robotics Engineer | Ocius Technologies | India | Feb 2020 – Aug 2020

- Contributed to the design and prototyping of the robotic leg, including aspects of mechanical modeling, motor selection, and joint optimization, which led to about 15% gains in mobility efficiency.
- Utilized Raspberry Pi for high-level system control and data processing
- Worked on the motion control algorithm design using C/C++ and Python and improved response time by 20%.
- Collaborated with multidisciplinary teams (mechanical, electrical, and software) to execute system-level testing and refinements
- Developed various scripts in Python for data logging, sensor validation, and motor calibration,

Embedded Systems Internship | Pantech Solutions | India | Feb 2024 – Aug 2024

- Embedded C programs were developed and optimized for ARM Cortex M4 and FPGA-based systems.
- Python was used for scripting automation tasks and for real-time sensor data analysis.
- Communication protocols such as UART, SPI, and I2C were implemented to interface external devices.
- Contributed to hardware/software integration by writing and testing firmware to ensure accurate communication between sensors and microcontrollers.

Education

Master of Computer Science

Vistula University | Poland
Mar 2023 - Sep 2024

Bachelor of Electronics and Communication Engineering

Anna University | India
Jul 2015 - Apr 2019